

Ziyue Zeng

Ph.D. Student, Waseda University

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Research Interests

Generative compression · Compressed representation understanding ·
Video generation and frame interpolation · Diffusion-based deepfake detection

Education

Ph.D. in Fundamental Science and Engineering Waseda University, Kato Laboratory	Sep 2026 – Present Tokyo, Japan
Research focus: generative compression and compressed representation understanding.	
M.S. in Fundamental Science and Engineering Waseda University, Watanabe Laboratory	Sep 2024 – Jul 2026 Tokyo, Japan
Exchange Student (Undergraduate) Waseda University	Apr 2023 – Sep 2023 Tokyo, Japan
B.S. in Artificial Intelligence Chongqing University, School of Big Data & Software Engineering	Sep 2020 – Jul 2024 Chongqing, China

Research Experience

Research Intern, CG Lab Huawei Tokyo Da Vinci Research Institute	Jun 2026 – Present Tokyo, Japan
Research Assistant National Institute of Information and Communications Technology (NICT)	Apr 2025 – Apr 2026 Tokyo, Japan
- Applied frame interpolation techniques to video compression and transmission. - Proposed Bi-AGMI, diffusion-based bidirectional keyframe interpolation (IEEE GCCE 2025, Oral Presentation Award). - Proposed FRS, motion-aware video segmentation with generative reconstruction (IEVC 2026).	
Graduate Researcher, Watanabe Laboratory Waseda University	Sep 2024 – Jul 2026 Tokyo, Japan
- Proposed GVCC, zero-shot video compression that uses a pretrained video generative model as the decoder (NeurIPS 2026), and GVCCTurbo, a rate-compute scheduler for it. - Proposed TSG, a universal synthesized image detector built on pretrained diffusion features (ACM MMAsia 2025, Oral). - Collaborated on large-motion frame interpolation with Stable Video Diffusion and ControlNet. - Collaborated with Yui Tatsumi on scalable image and video coding for humans and machines (IEEE MMSP 2025, IEEE GCCE 2025, IEEE ICCE 2026).	
Research Assistant, Information Processing Lab Chongqing University	Sep 2021 – Jun 2024 Chongqing, China
Research on evidence theory and information fusion; two first-author journal articles.	

Publications

Conference Papers

- [1] **Ziyue Zeng**, Xun Su, Haoyuan Liu, Bingyu Lu, Yui Tatsumi, Hiroshi Watanabe. **GVCC: Zero-Shot Video Compression via Codebook-Driven Stochastic Rectified Flow**. *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. To appear.
- [2] **Ziyue Zeng**, Yui Tatsumi, Hiroshi Watanabe. **Flow Residual Segmentation and Generative Reconstruction for Motion-Aware Video Coding**. *IEEE International Conference on Image Electronics and Visual Computing (IEVC)*, 2026.
- [3] Yui Tatsumi, **Ziyue Zeng**, Hiroshi Watanabe. **Training-Free Adaptive Quantization for Variable Rate Image Coding for Machines**. *IEEE International Conference on Consumer Electronics (ICCE)*, 2026.
- [4] **Ziyue Zeng**, Yupei Guo, Haoyuan Liu, Dingjie Peng, Hiroshi Watanabe. **Time Step Generating: A Universal Synthesized Deepfake Image Detector**. *ACM International Conference on Multimedia in Asia (MMAsia)*, 2025. Oral.

- [5] **Ziyue Zeng**, Yui Tatsumi, Hiroshi Watanabe. [Bidirectional Attention-Gated Motion Injection for Frame Interpolation](#). *IEEE Global Conference on Consumer Electronics (GCCE)*, 2025. **Oral Presentation Award**.
- [6] Yui Tatsumi, **Ziyue Zeng**, Hiroshi Watanabe. [Seed Selection for Human-Oriented Image Reconstruction via Guided Diffusion](#). *IEEE Global Conference on Consumer Electronics (GCCE)*, 2025. **Oral Presentation Award**.
- [7] Yui Tatsumi, **Ziyue Zeng**, Hiroshi Watanabe. [Explicit Residual-Based Scalable Image Coding for Humans and Machines](#). *IEEE International Workshop on Multimedia Signal Processing (MMSP)*, 2025.

Journal Articles

- [8] **Ziyue Zeng**, Fuyuan Xiao. [A Generalized Hellinger Distance for Multisource Information Fusion and Its Application in Pattern Classification](#). *Computational and Applied Mathematics*, 43(1), 2024.
- [9] **Ziyue Zeng**, Fuyuan Xiao. [A New Complex Belief Entropy of \$\chi^2\$ Divergence with Its Application in Cardiac Interbeat Interval Time Series Analysis](#). *Chaos, Solitons & Fractals*, 172:113542, 2023.

Preprints

- [10] **Ziyue Zeng**, Dingjie Peng, Xun Su, Hiroshi Watanabe. [GVCCTurbo: Rate-Compute Quality Scheduling for Codebook Driven Generative Compression](#). *arXiv:2608.03517*, 2026.

Domestic Conferences (Japan)

- [11] Yui Tatsumi, **Ziyue Zeng**, Hiroshi Watanabe. Scalable Variable-Rate Video Compression for Humans and Machines via Keyframe Interpolation (in Japanese). *Meeting on Image Recognition and Understanding (MIRU)*, 2026. **Interactive Presentation Award**.
- [12] Yui Tatsumi, **Ziyue Zeng**, Hiroshi Watanabe. Assessing the Effectiveness of Residual Information in Scalable Image Coding for Humans and Machines (in Japanese). *Meeting on Image Recognition and Understanding (MIRU)*, 2025.

Awards

IEEE GCCE 2025 Oral Presentation Award
Bi-AGMI (first author)

2025

Skills

- **Programming:** Python, PyTorch, Diffusers
- **Generative models:** fine-tuning and inference of video diffusion and rectified-flow models (SVD, Wan)
- **Writing:** LaTeX, scientific plotting